WESTMINSTER
International University in Tashkent

Module name and code	**Systems Analysis and Design 6BUIS026C**
CW number and weighting	CW1 - 40%
Contact details and office hours of the lecturer setting the task	Said Abduvaliev sabduvaliev@wiut.uz, Fridays 11:00-12:00
Submission deadline	26-10-2026
Results date and type of feedback	09-11-2026
The CW checks the following learning outcomes:	
1. LO1 - Apply a variety of techniques for requirements elicitation (for well-defined problems medium complexity and restricted scope). 2. LO2 - Identify, document, and prioritise functional requirements of medium complexity. 3. LO3 - Model the main data that is needed to support a given set of functional requirements.	

Task

Overview

This is a group coursework. You will analyse an existing local business and document your findings. You will have to choose a correct data and requirements collection techniques, justify them and then document your findings using the standard notations and diagrams.

The main goal of the assessment is to build skills in analysing existing business processes, suggest optimal solutions and document the analysis stage in standard way. Additionally, this will help you to develop skills that would allow a better context building and getting better results from AI agents when building a complex IT system.

All works must be submitted through WIUT Intranet's online assessment submission functionality.

Description

This is a group assessment. You must form into groups of 4. It is your responsibility to form groups. Please note that you must form a group of exactly 4 students (groups of 3 or 5 are not allowed). If you cannot find a group, add your student ID and contact details to the first empty row on this Spreadsheet. Contact your groupmates.

As a group, you will select an existing local company. It can be a small grocery shop near your apartment or a well-known large organization in Uzbekistan. You are not allowed to select some virtual enterprise or non-local company. Failing to select a local company will result in 0 points.

You will need to carry out analysis of the business and document any existing system(s). Optionally, you can also identify a business issue and suggest a new solution. You can document an existing system or proposed solutions for a new system that would solve the business' problem.

Requirements

1. **Business case (10 points):** Provide a short business and business processes description to give a general idea about the company. It can be similar to the cases we have used during our tutorials (Threadz, Fresco, etc.). The section cannot be longer than 500 words. You are allowed to use generative AI (e.g. ChatGPT). However, make sure that each sentence conveys an information. If general text with no important information is identified, you will be penalised.

2. **Requirements gathering process (15 points):** Conduct a requirements gathering process. You may use interview, observation, or any other methods you feel are appropriate. You need to justify your choice of data collection. You must collect at least some of your data from prospective users. Record your data carefully – and make sure you can provide evidence of carrying out data gathering processes in your report (e.g. audio recording or transcript of your interview, filled out surveys, etc.).

3. **Use Case Diagram (10 points):** Based on your findings, draw a Use Case Diagram.

4. **Use Cases (15 points):** Take any four Use Cases from your Use Case Diagram and produce 4 Use Cases (table with the Use Case details, see Tutorial 3).

5. **Data Flow Diagram (10 points):** Based on your findings, draw a Data Flow Diagram. Draw all level 0 and level 1 diagrams. Optionally, you can draw level 2 diagram.

6. **Activity Diagram (10 points):** Produce Activity Diagram for one complex business process. If your business case does not have complex business processes, you can produce diagrams for several small processes.

7. **Sequence diagram (10 points):** Produce Sequence Diagram for four main functions/processes.

8. **Requirements (20 points):** Based on the gathered data and your subsequent interpretation and analysis of the data, draw requirements catalogue summaries and identify five functional and five non-functional requirements for the information system. Please use the format given below for documenting the requirements.

Requirements ID	Requirements name	Requirement Description

You may use any diagramming tools to draw the diagrams. There is no work count limitation.

Your report should be analytical. Please do not explain any of the concepts. Descriptive content could be used in the Business case section. You will not be penalized for a short and concise report.

Deliverables

Report in Microsoft Word format containing:

1. Business case.
2. Justification of the chosen requirements gathering methods and an evidence of carrying out the requirements gathering process.
3. Use case diagram.
4. Four Use cases.
5. Data flow diagram.
6. Activity diagram.
7. Sequence diagram.
8. Requirements catalogue.

Submission

Please name your file in the following format: **`SAD_CW1_STUDENTID.docx.`** For example: `SAD_CW1_1234.docx`

Your report must be in Microsoft Word file format. Do not archive (zip) your file.

Submit your file to “Coursework 1” assessment on WIUT Intranet.

Format

- Word-processed Arial 12, double-spaced.
- Do not exceed the number of words. Refer to the tariffs available on Registrar’s portal.
- Include a content page with each section's headings and page numbers if appropriate. Pages should be numbered.
- Use the Westminster-Harvard (Oscola for Law) Style for Referencing.

Assessment criteria/marking scheme

Criteria	Mark
Business case The description of the business includes: • The business name and its industry: 1 point. • The main activities of the business: 2 points. • Detailed description of the core/main business processes: 7 points. • The text is not concise, and it carries no information: -5points.	10
Requirements gathering process • 2 requirements gathering methods are clearly stated: 3 points. • The chosen methods are well justified: 5 points. • The carried-out requirements gathering is done effectively: 5 points. • Considerable effort was put into the process: 2 points. • No evidence of conducting requirements gathering: -10 points.	15
Use Case Diagram • Use case diagram is drawn according to the standard: 3 points. • All main actors and processes are documented: 4 points. • The diagram uses <<includes>> and <<extends>>: 3 points. • The diagram lacks obvious actors, use cases: -3 points.	10
Use Cases (15 points) • The use cases are using the standard notation: 5 points. • There are at least 4 use cases: 4 points. • The use cases are detailed and good quality: 6 points. • The use cases are missing “Extensions” section: -4 points.	15
Data Flow Diagram (10 points) • The diagram has at least one level 0 & tree level 1 diagrams: 5 points. • The diagram documents correct data flow fully: 5 points. • The diagram doesn’t use the standard notation: -5 points.	10
Activity Diagram • The diagram is documenting a complex process correctly: 6 points. • The diagram is using all shapes correctly: 4 points. • The diagram doesn’t use the standard notation: -5 points.	10
Sequence diagram • There are four processes documented correctly: 4 points. • The diagram is using all shapes correctly: 6 points. • The diagram doesn’t use the standard notation: -5 points.	10
Requirements • There are 5 functional and 5 non-functional requirements: 8 points. • All requirements are detailed and cover all edge cases: 12 points. • Requirements are short and don’t cover all the details: -2 points/req.	20